

- a. All 4 worked
- b. All 4
- c. I learned how to create and manipulate arrays of 1 dimension in multiple ways.
Specifically I became much more comfortable with using the `arange` function to create arrays, as well using element by element wise operations to modify them. I also became more comfortable with iterating through arrays. Finally I learned how to read text files and get data from them, and then manipulate that data.

Code:

```
import math
```

```
import numpy as np
```

```
#STEP 1
```

```
A3 = np.array([[3,4,5],[4,5,6]])
```

```
A2 = np.array([[1,2,3,4],[4,3,2,1],[0,9,8,7]])
```

```
A1 = np.array([2.0, 3, 4, 5])
```

```
def step1_a():
```

```
    print('Add constant:', A1 + 4)
```

```
def step1_b():
```

```
    print('Multiply constant:', A1 * 4)
```

```
def step1_c():
```

```
    print('Multiply by itself:', A1 * A1)
```

```
    #print(A1**2) does the same
```

```
def step1_d():
```

```
    print('Add two arrays:\n', A1 + A2)
```

```
    #print(A1 + A3) does not work
```

```
def step1_e():
```

```
    print('Multiply two arrays:\n', A1 * A2)
```

```
    #print(A2 * A3) does not work
```

```
#step1_a(), step1_b(), step1_c(), step1_d(), step1_e()
```

```
#STEP 2
```

```
def step2_method1():
```

```
    """Start with empty arrays, and then append each value for both x and y"""
```

```
    x = np.array([])
```

```
y = np.array([])
```

```
for i in np.arange(0,10.01,0.01):
```

```
    x = np.append(x,i)
```

```
    y = np.append(y,np.sin(i))
```

```
print('Method 1:\n', x,y)
```

```
def step2_method2():
```

```
    """Create the x array directly and add each y value one by one"""
```

```
    x = np.arange(0,10.01,0.01)
```

```
    y = ([])
```

```
    for i in x:
```

```
        y = np.append(y,np.sin(i))
```

```
    print('Method 2:\n', x,y)
```

```
def step2_method3():
```

```
    """Create the x array directly and change the entire array to make the y array directly"""
```

```
    x = np.arange(0,10.01,0.01)
```

```
    y = np.sin(x)
```

```
    print('Method 3:\n', x, y)
```

```
#step2_method1(), step2_method2(), step2_method3()
```

```
#STEP 3
```

```
#I can call functions from the console or have them called from the script
```

```
#STEP 4
```

```
def step4_1(array = A1):
```

```
    total = sum(array)
```

```
    length = len(array)
```

```
    average = total / length
```

```
    print('1:', average)
```

```
def step4_2(array = A1, C = 2):
```

```
    total = sum(array)
```

```
    length = len(array)
```

```
    average = total / length
```

```
    average = average * C
```

```
    print('2:', average)
```

```
def step4_3():
```

```
    row_1, row_2 = np.loadtxt('Lab_1.txt')
```

```
total_1 = sum(row_1)
length_1 = len(row_1)
average_1 = total_1 / length_1
print('3 Row 1:', average_1)
```

```
total_2 = sum(row_2)
length_2 = len(row_2)
average_2 = total_2 / length_2
print('3 Row 2:', average_2)
```

```
def step4_function(x):
    return 4*x**2 + 7*x + 3
```

```
def step4_4():
    y = step4_function(A1)
    total = sum(y)
    length = len(y)
    average = total / length
    print('4:', average)
```

```
#step4_1(), step4_2(), step4_3(), step4_4()
```